

S32K1xxDSAD

Addendum For S32K1xx

Rev. 1 — 3 February 2026

Data sheet addendum
CONFIDENTIAL

1 Introduction

This document is created in support of selected S32K11x and S32K14x derivative devices within the S32K1 automotive microcontroller family.

This document must be used in conjunction with the latest version of the S32K1xx Data Sheet available on <http://www.nxp.com>

S32K116V, S32K118V, S32K142N, and S32K144N devices are specified for operation with a 3.3 V VDD supply only. Operation with a 5 V VDD supply is not supported for these devices. These devices are members of the S32K11x and S32K14x product families and are supported by NXP's long-term product longevity program.

S32K116V, S32K118V, S32K142N, and S32K144N devices are derivatives of the corresponding standard S32K11x and S32K14x products and provide a reduced and well-defined subset of functionality. These devices implement the same underlying feature set; however, functionality is tested and guaranteed only within the limits, configurations, and capabilities explicitly specified for these devices. Feature availability and supported operating limits are defined in the applicable [Feature Comparison](#). Operation beyond the specifications explicitly stated is not guaranteed.

2 Feature comparison

Table 1. Feature Comparison

	S32K11x		S32K14x	
Parameter	S32K116V	S32K118V	S32K142N	S32K144N
Flash	128 KB	256 KB	256 KB	512 KB
SRAM	17 KB	25 KB	32 KB	64 KB
Data-Flash	Up to 32 KB (2 KB Flex RAM)	Up to 32 KB (2 KB Flex RAM)	Up to 64 KB (4 KB Flex RAM)	Up to 64 KB (4 KB Flex RAM)
Core	Cortex M0+	Cortex M0+	Cortex M4F	Cortex M4F
Frequency	48 MHz	48 MHz	80 MHz	80 MHz
VDD	2.7 – 3.6 V	2.7 – 3.6 V	2.7 – 3.6 V	2.7 – 3.6 V
eDMA	4ch	4ch	8ch	8ch
CAN FD (CAN)	1 (1)	1 (1)	1 (1)	1 (1)
UART / LIN	2	2	2	2
SPI	1	1	1	1
I2C	1	1	1	1
FlexIO	1	1	1	1
ADC (12-bit)	1x (13ch)	1x (16ch)	1x (16ch)	1x (16ch)



Table 1. Feature Comparison...continued

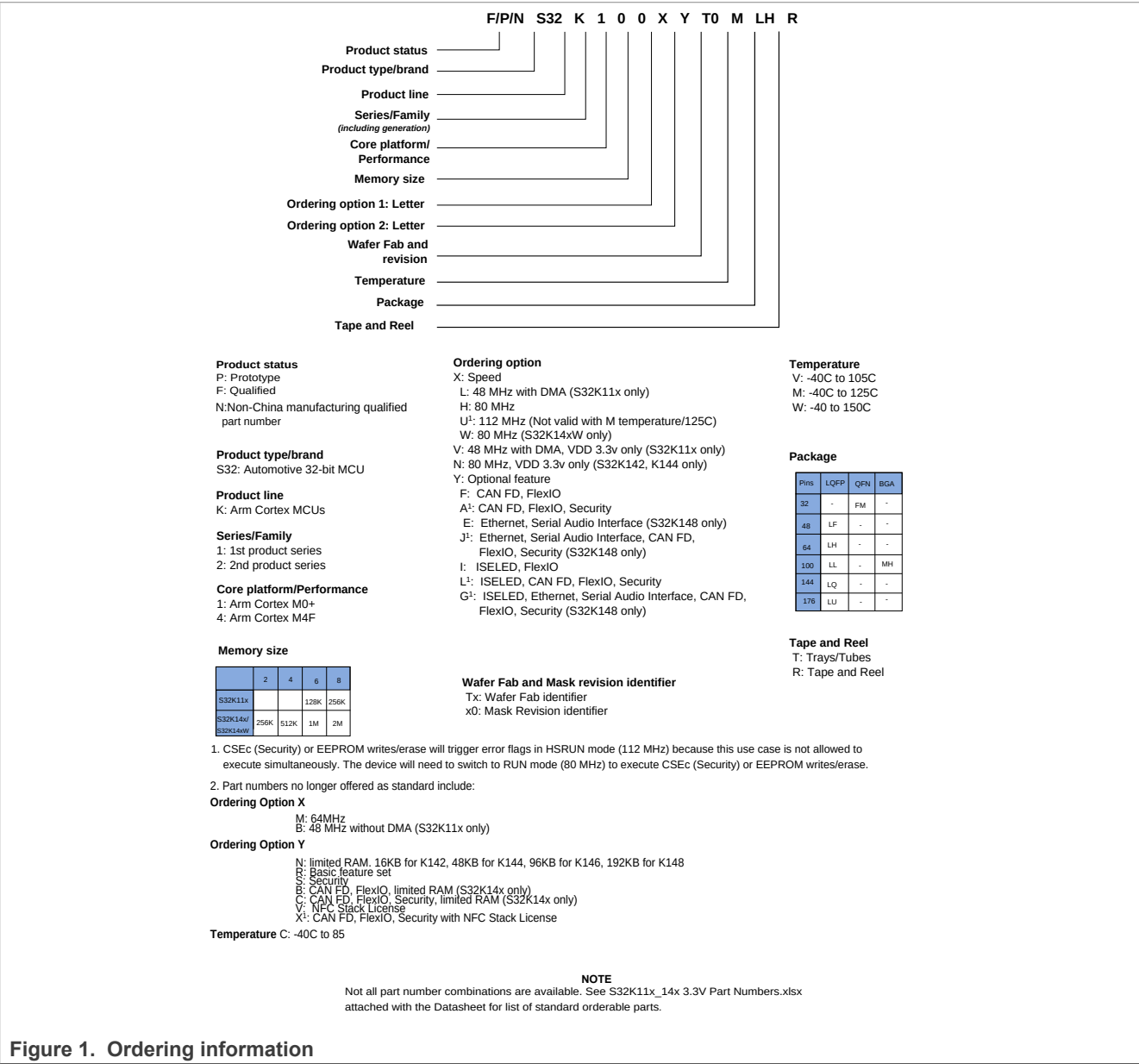
	S32K11x		S32K14x	
Analog Comparator	1	1	1	1

3 Ordering information

Not all part number combinations are available. See the attachment S32K11x_14x 3.3V Part Numbers.xlsx attached with the Datasheet for a list of standard orderable part numbers.

Note: Bridge Solution Part numbers (Ordering Option X):

- V: 48 MHz with DMA, VDD 3.3v only (S32K11x only)
- N: 80 MHz, VDD 3.3v only (S32K142, K144 only)



4 Obtaining package dimensions

Package dimensions are provided in the package drawings.

To find a package drawing, go to <http://www.nxp.com> and perform a keyword search for the drawing's document number:

Package option	Document Number	Manufacture Code
32-pin QFN	SOT617-3 ^[1]	98ASA01350D
48-pin LQFP	SOT313-3	98ASH00962A
64-pin LQFP	SOT1699-1	98ASS23234W
100-pin LQFP	SOT407-3	98ASS23308W

[1] 5x5 mm package

5 DC electrical specifications at 3.3 V Range

Note: For details on the pad types defined in [Table 2](#), see Reference Manual section IO Signal Table and IO Signal Description Input Multiplexing sheet(s) attached with Reference Manual.

Table 2. DC electrical specifications at 3.3 V Range for S32K1xx series

Symbol	Parameter	Value			Unit	Notes
		Min.	Typ.	Max.		
V_{DD}	I/O Supply Voltage	2.7	3.3	4	V	
V_{ih}	Input Buffer High Voltage	$0.7 \times V_{DD}$	—	$V_{DD} + 0.3$	V	[1]
V_{il}	Input Buffer Low Voltage	$V_{SS} - 0.3$	—	$0.3 \times V_{DD}$	V	[2]
V_{hys}	Input Buffer Hysteresis	$0.06 \times V_{DD}$	—	—	V	
I_{ohGPIO} $I_{ohGPIO-HD_DSE_0}$	I/O current source capability measured when pad $V_{oh} = (V_{DD} - 0.8 \text{ V})$	3.5	—	—	mA	
I_{olGPIO} $I_{olGPIO-HD_DSE_0}$	I/O current sink capability measured when pad $V_{ol} = 0.8 \text{ V}$	3	—	—	mA	
$I_{ohGPIO-HD_DSE_1}$	I/O current source capability measured when pad $V_{oh} = (V_{DD} - 0.8 \text{ V})$	14	—	—	mA	[3]
$I_{olGPIO-HD_DSE_1}$	I/O current sink capability measured when pad $V_{ol} = 0.8 \text{ V}$	12	—	—	mA	[3]
$I_{ohGPIO-FAST_DSE_0}$	I/O current sink capability measured when pad $V_{oh} = V_{DD} - 0.8 \text{ V}$	9.5	—	—	mA	[4]
$I_{olGPIO-FAST_DSE_0}$	I/O current sink capability measured when pad $V_{ol} = 0.8 \text{ V}$	10	—	—	mA	[4]
$I_{ohGPIO-FAST_DSE_1}$	I/O current sink capability measured when pad $V_{oh} = V_{DD} - 0.8 \text{ V}$	16	—	—	mA	[4]
$I_{olGPIO-FAST_DSE_1}$	I/O current sink capability measured when pad $V_{ol} = 0.8 \text{ V}$	15.5	—	—	mA	[4]
IOHT	Output high current total for all ports	—	—	100	mA	
IIN	Input leakage current (per pin) for full temperature range at $V_{DD} = 3.3 \text{ V}$	—	5	$500^{[5]}$	nA	[6]

Table 2. DC electrical specifications at 3.3 V Range for S32K1xx series...continued

Symbol	Parameter	Value			Unit	Notes
		Min.	Typ.	Max.		
	All pins other than high drive port pins		0.005	0.5	μA	
	High drive port pins		0.010	0.5	μA	
R _{PU}	Internal pullup resistors	20		60	kΩ	[7]
R _{PD}	Internal pulldown resistors	20		60	kΩ	[8]

[1] For reset pads, same V_{ih} levels are applicable[2] For reset pads, same V_{il} levels are applicable

[3] The value given is measured at high drive strength mode. For value at low drive strength mode see the Ioh_Standard value given above.

[4] For reference only. Run simulations with the IBIS model and custom board for accurate results.

[5] Typical leakage is given at room temperature. Maximum is given for 125°C. Leakage numbers increase with temperature, approximately every 12 – 14°C the value doubles. Leakage is tested at hot temperature. We ensure maximums are not exceeded. Please note that when the ADC module samples a pin, additional currents beyond the leakage number are drawn to charge the sample and hold capacitances and internal analog busses. These are difficult to predict

[6] Several I/O have both high drive and normal drive capability selected by the associated Portx_PCRn[DSE] control bit. All other GPIOs are normal drive only. For details see IO Signal Description Input Multiplexing sheet(s) attached with the *Reference Manual*.[7] Measured at input V = V_{SS}[8] Measured at input V = V_{DD}

6 AC electrical specifications at 3.3 V range

Table 3. AC electrical specifications at 3.3 V Range for S32K1xx series

Symbol	DSE	Rise time (nS) ^[1]		Fall time (nS) ^[1]		Capacitance (pF) ^[2]
		Min.	Max.	Min.	Max.	
tRF _{GPIO}	NA	3.2	14.5	3.4	15.7	25
		5.7	23.7	6.0	26.2	50
		20.0	80.0	20.8	88.4	200
tRF _{GPIO-HD}	0	3.2	14.5	3.4	15.7	25
		5.7	23.7	6.0	26.2	50
		20.0	80.0	20.8	88.4	200
	1	1.5	5.8	1.7	6.1	25
		2.4	8.0	2.6	8.3	50
		6.3	22.0	6.0	23.8	200
tRF _{GPIO-FAST}	0	0.6	2.8	0.5	2.8	25
		3.0	7.1	2.6	7.5	50
		12.0	27.0	10.3	26.8	200
	1	0.4	1.3	0.38	1.3	25
		1.5	3.8	1.4	3.9	50
		7.4	14.9	7.0	15.3	200

[1] For reference only. Run simulations with the IBIS model and your custom board for accurate results.

[2] Maximum capacitances supported on Standard IOs. However interface or protocol specific specifications might be different, for example for ENET, QSPI etc. . For protocol specific AC specifications, see respective sections.

7 Revision History

The following table provides a revision history for this document.

Table 4. Revision History

Document ID	Release date	Description
S32K1DSAD v.1	28 January 2026	Initial release

Legal information

Data sheet status

Document status ^{[1][2]}	Product status ^[3]	Definition
Objective [short] data sheet	Development	This document contains data from the objective specification for product development.
Preliminary [short] data sheet	Qualification	This document contains data from the preliminary specification.
Product [short] data sheet	Production	This document contains the product specification.

[1] Please consult the most recently issued document before initiating or completing a design.

[2] The term 'short data sheet' is explained in section "Definitions".

[3] The product status of device(s) described in this document may have changed since this document was published and may differ in case of multiple devices. The latest product status information is available on the Internet at URL <https://www.nxp.com>.

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Contents

1 Introduction 1

2 Feature comparison 1

3 Ordering information 2

4 Obtaining package dimensions 3

5 DC electrical specifications at 3.3 V
Range 3

6 AC electrical specifications at 3.3 V
range 4

7 Revision History 5

Legal information 6

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